# Latent Space Behavior in Self-Supervised Visual Learning:

## *A Comparative Study of Contrastive and Reconstruction-Based Representation Learning*

*AIMS Research Internship - Representational Learning Task- Round 2*

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# Abstract


Self-Supervised Learning (SSL) is a type of machine learning approach where a model is trained using data that does not have labels or answers provided. Instead of needing human supervised labelling of data, the model finds patterns and creates its own labels from data by itself. SSL is useful in situations pertaining incomplete labelling of data and when labelling data is manually time consuming and expensive. SSL is essentially a middle ground between supervised learning, (with labels) and unsupervised learning (without labels). SSL makes it easier to adapt models to new tasks by using the knowledge gained from pre-training on unlabeled data.

This study investigates how different learning objectives influence latent representation geometry, semantic consistency and contextual feature preservation. Latent embeddings extracted from trained models were analyzed using various visualization and reconstruction techniques.

Experimental observations indicate that contrastive learning produced more discriminative and locally compact latent representations, while reconstruction-based learning preserved smoother structural continuity and contextual similarity. Despite lightweight architectures and constrained computational resources, the experiments demonstrate clear objective-dependent differences in learned embedding behavior. The findings highlight the tradeoff between discriminative separation and contextual representation learning in modern and legacy self-supervised learning frameworks.

Latent embeddings extracted from the improved models were analyzed through t-SNE and UMAP dimensionality reduction, 3D interactive UMAP visualization, k-nearest-neighbor (k-NN) linear evaluation, SVD eigenvalue spectrum analysis, intraclass versus interclass similarity profiling, MAE reconstruction quality assessment, and gradient-based saliency and activation mapping. The results demonstrate that SimCLR produces discriminative but semantically compact clusters with strong augmentation invariance, whereas MAE develops contextually coherent, structurally smooth representations which were suited to spatial reconstruction. Interestingly, extended training duration was found to be a decisive factor in representation quality for both approaches, with baseline short-epoch runs producing qualitatively inferior embeddings.

These findings underscore the importance of matching the SSL objective to the intended downstream task and highlight the value of sufficient training in realizing the representational potential of each method.

# 1. Introduction

## 1.1 Representation Learning

A *representation* is a mapping from raw data to a more compact, structured encoding that captures the essential information while discarding irrelevant detail

The quality of a learned representation strongly influences the performance of downstream tasks such as image classification, object detection, retrieval, segmentation, and clustering. Effective latent spaces preserve semantic similarity, meaning that visually or conceptually similar inputs are mapped to nearby regions in the embedding space. This property enables models to generalize more effectively across different tasks and datasets.

## 1.2 Self-Supervised Learning

*Self-supervised learning* uses the data itself to generate a training signal. Instead of relying on manual labels, the model constructs a predictive task from the input distribution and learns a transferable representation from that surrogate objective.

This is especially useful when annotations are expensive, incomplete, or unavailable. The learned objective can be contrastive, predictive, or generative, but the common goal is to extract structure from unlabeled images.

## 1.3 Motivation

Although SimCLR and MAE have achieved strong results on large-scale benchmarks, their behavior in more constrained settings remains less well understood. Many practical applications operate with smaller datasets, limited computational resources, and shorter training schedules than those typically reported in the contemporary ML literature. As a result, understanding how representation quality evolves under such constraints is an important research question.

In this study, an obstacle detection dataset is used as a representative computer vision task to investigate how training duration influences learned representations. The analysis compares both methods across shorter baseline runs and longer, more thoroughly trained configurations, allowing the impact of additional training to be examined directly through changes in latent-space structure and feature quality.

A second motivation of this work is methodological. Rather than relying on a single evaluation metric, the learned representations are examined through multiple complementary perspectives. These include dimensionality reduction techniques, downstream classification performance, spectral analysis of embeddings, reconstruction quality assessment, and saliency-based visualization. Together, these analyses provide a broader understanding of how different self-supervised learning objectives shape the representations learned by SimCLR and MAE.

## 1.4 Project Objectives

The objective of this project is to investigate how different self-supervised learning objectives influence the structure and behavior of learned latent representations in visual embedding spaces. Two representative SSL paradigms were selected for comparison: SimCLR, a contrastive learning framework, and Masked Autoencoders (MAE), a reconstruction-based framework.

To analyze the learned embeddings, multiple evaluation techniques are employed, including dimensionality reduction using t-SNE and UMAP, clustering metrics, reconstruction analysis, and augmentation consistency testing. The experiments are conducted on an obstacle detection dataset containing approximately 19,000 images, with embeddings extracted primarily from the training split.

This work is not intended to be an extensive large-scale benchmark study and due to limited available time and computational resources, the experiments were performed using reduced training schedules (around 10 epochs), lightweight backbone architectures, and a constrained evaluation scope. Consequently, the results should be interpreted primarily as an exploratory investigation into latent space behavior under practical resource limitations rather than as a fully optimized comparison of self-supervised learning methods.

# 2. Theoretical Background

## 2.1 SimCLR

Contrastive learning methods aim to bring representations of similar images closer while pushing dissimilar images apart in embedding space. SimCLR (*Simple Framework for Contrastive Learning of Visual Representations*) achieves this by creating two augmented views of the same image and treating them as a positive pair. The model is trained using the NT-Xent contrastive loss, which maximizes similarity between positive pairs while separating embeddings of other samples in the batch.

As a result, SimCLR typically produces discriminative latent spaces with compact semantic clusters and strong augmentation invariance.

The NT-Xent (Normalized Temperature-scaled Cross-Entropy) loss encourages the cosine similarity between positive pairs to approach one while pushing the representations of all other images in the batch apart. Formally, for a positive pair (i, j), the loss is:

$$l(i,j) = -\log \left[ \exp(\mathrm{sim}(z_i,z_j)/\tau) \,/\, \Sigma_{k \neq i} \exp(\mathrm{sim}(z_i,z_k)/\tau) \right]$$

where sim(·,·) denotes cosine similarity, $\tau$ is a temperature hyperparameter, and the sum is taken over all $2N - 1$ other samples in the batch. After pretraining, representations are extracted from the encoder prior to the projection head, as this layer is known to retain more semantic information.

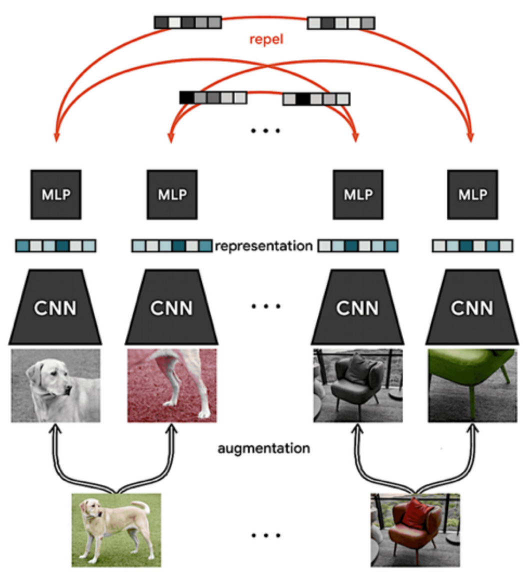


*Figure 1. Schematic of the SimCLR contrastive learning pipeline. Two stochastic augmentations of the same image constitute a positive pair. The shared ResNet-50 encoder and projection head produce embeddings that are trained with NT-Xent loss to be augmentation-invariant.*

## 2.2 Masked Autoencoders (MAE)

Masked Autoencoders (MAE) follow a reconstruction-based learning strategy. A large portion of image patches is randomly masked, and the model learns to reconstruct the missing regions from the visible patches. The encoder processes only unmasked patches, while a lightweight decoder predicts the masked content using a pixel reconstruction loss.

This objective encourages the model to learn contextual and structural relationships within images rather than strong instance-level separation.

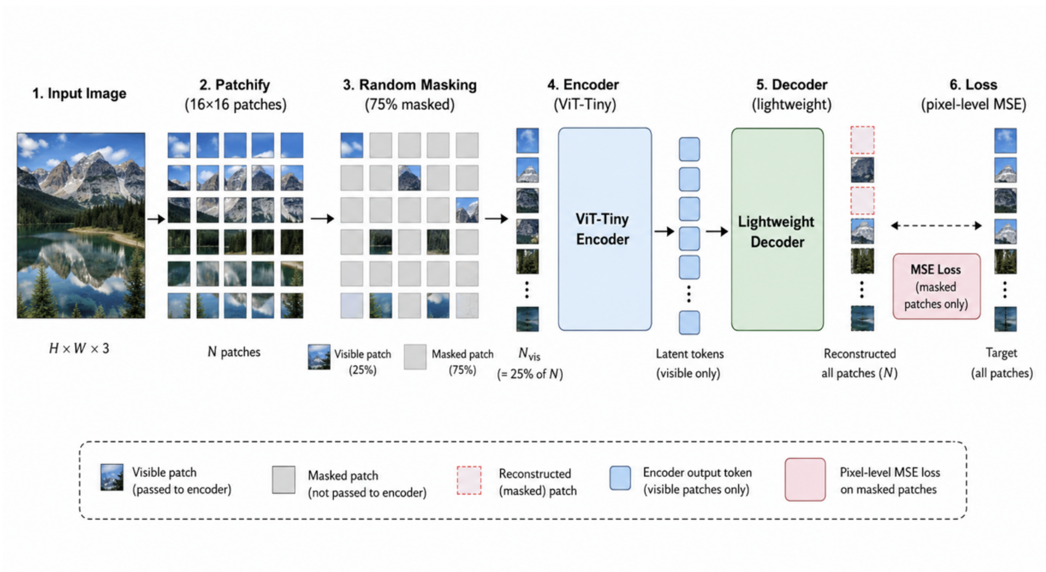


*Figure 2. Schematic of the Masked Autoencoder (MAE) pipeline. A random 75% of image patches are masked prior to encoding. The ViT-Tiny encoder processes only visible patches, and a lightweight decoder reconstructs the masked regions using pixel-level MSE loss.*

# 3. Dataset and Preprocessing

## 3.1 Dataset Description

The dataset used was an objection detection dataset by Kaggle. A small dataset by usual research standards. It contained sufficient images for a low resource model training intended solely for exploratory investigation into behavior of simCLR and MAE models.

The dataset contains significant visual variation, including changes in scale, viewpoint, illumination, and partial occlusion. These characteristics make the dataset suitable for evaluating the ability of self-supervised learning models to capture robust semantic and structural representations without explicit labels.

Only the training split was primarily used for latent representation analysis since the models were trained exclusively on this subset and had not fully converged under the limited training schedule.

- Kaggle Object Detection Dataset 24k images with ~19k images used for training
- Challenges include occlusion, scale variation, and visual diversity
- Experiments performed under limited compute for exploratory analysis

The dataset presents a realistic and challenging distribution for self-supervised evaluation. The significant intraclass variation in some instance showed objects appearing at different scales, distances, and under varying illumination. It pushed SSL methods to form semantically coherent clusters. At the same time, the breadth of obstacle categories provides a sufficiently rich label space to serve as a meaningful evaluation signal for downstream k-NN classification.

It is important to note that no ground-truth labels were used during pretraining. Labels were reserved exclusively for k-NN evaluation and qualitative verification of cluster semantic content, in keeping with the self-supervised paradigm.

## 3.2 Image Preprocessing

All images were resized to a fixed spatial resolution of 224 × 224 pixels before training. The images were then converted into tensor format and normalized using ImageNet mean and standard deviation values in order to maintain compatibility with pretrained computer vision architectures and stabilize optimization during training.

The preprocessing pipeline was intentionally kept lightweight and consistent across both SimCLR and MAE experiments to ensure fair comparison between the two self-supervised learning paradigms.

A uniform preprocessing pipeline was applied to all images before training under both paradigms to ensure a fair comparison. Images were resized to a fixed spatial resolution of 224 × 224 pixels and converted to PyTorch tensor format. Pixel values were normalized using ImageNet channel-wise mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]) statistics, a standard practice that stabilizes gradient flow and maintains compatibility with ResNet-50 pretrained initialization conventions

## 3.3 SimCLR Augmentation Pipeline

*Figure 3. Illustration of the SimCLR augmentation applied to representative training images. Each row displays two original images and next row shows each of its two independently augmented views used as positive pairs during contrastive training.*

The augmentation pipeline plays a foundational role in SimCLR, as it defines the set of transformations to which representations must be invariant. For each training image, two independent augmentations were sampled from the following sequence: random resized crop (scale range 0.2 to 1.0, ratio range 0.75 to 1.33), random horizontal flip with probability 0.5, color jitter applied with probability 0.8 (brightness ±0.4, contrast ±0.4, saturation ±0.4, hue ±0.1), random grayscale conversion with probability 0.2, and Gaussian blur with kernel size 23 and sigma uniformly sampled from [0.1, 2.0]. These augmentations collectively encourage the encoder to discard low-level nuisance factors while preserving semantic content.

## 3.4 MAE Masking Strategy

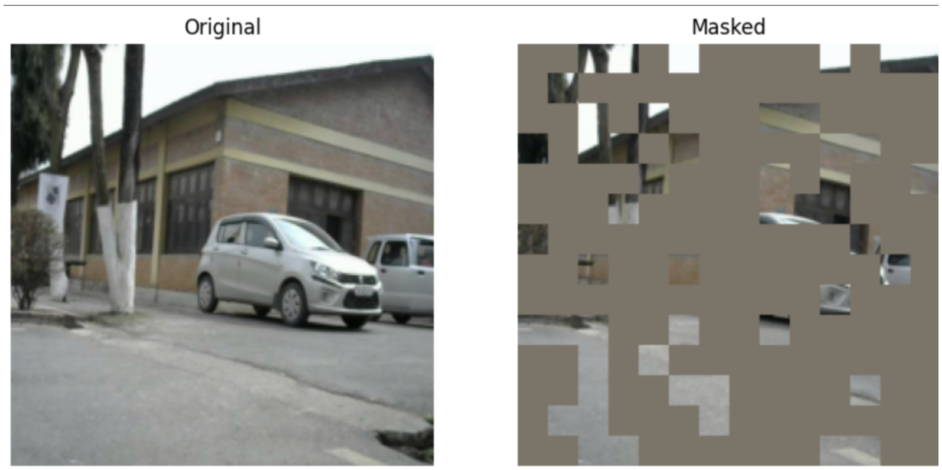


*Figure: Illustration of MAE augmentation. 75% of image was masked*

For MAE training, image preprocessing was intentionally minimal. Each image was resized to 224 × 224 pixels, normalized using ImageNet statistics, and partitioned into non-overlapping 16 × 16 pixel patches. This resulted in a 14 × 14 grid containing 196 patches per image.

Following the MAE training paradigm, a masking ratio of 75% was applied at every training iteration. Consequently, 147 patches were randomly masked, while only the remaining 49 visible patches (25% of the image) were provided as input to the encoder. The encoder was therefore required to learn compact latent representations from highly incomplete visual information. A lightweight decoder subsequently used these latent representations to reconstruct the missing image regions.

The 75% masking ratio was maintained consistently across all MAE experiments. This relatively aggressive masking strategy, recommended in the original MAE framework, discourages trivial pixel interpolation and encourages the model to learn higher-level semantic and structural representations rather than relying on local texture continuity alone.

# 4. Methodology

## 4.1 Baseline Exploratory Experiments

Prior to the main experiments, baseline training runs were conducted to establish reference behavior and identify the limitations of short-epoch training. SimCLR was trained with a ResNet-18 backbone for 10 epochs, and MAE was trained with a ViT-Tiny encoder for 10 epochs. Both configurations used the same data preprocessing and augmentation strategies described above, but with reduced training duration and smaller backbone capacity.

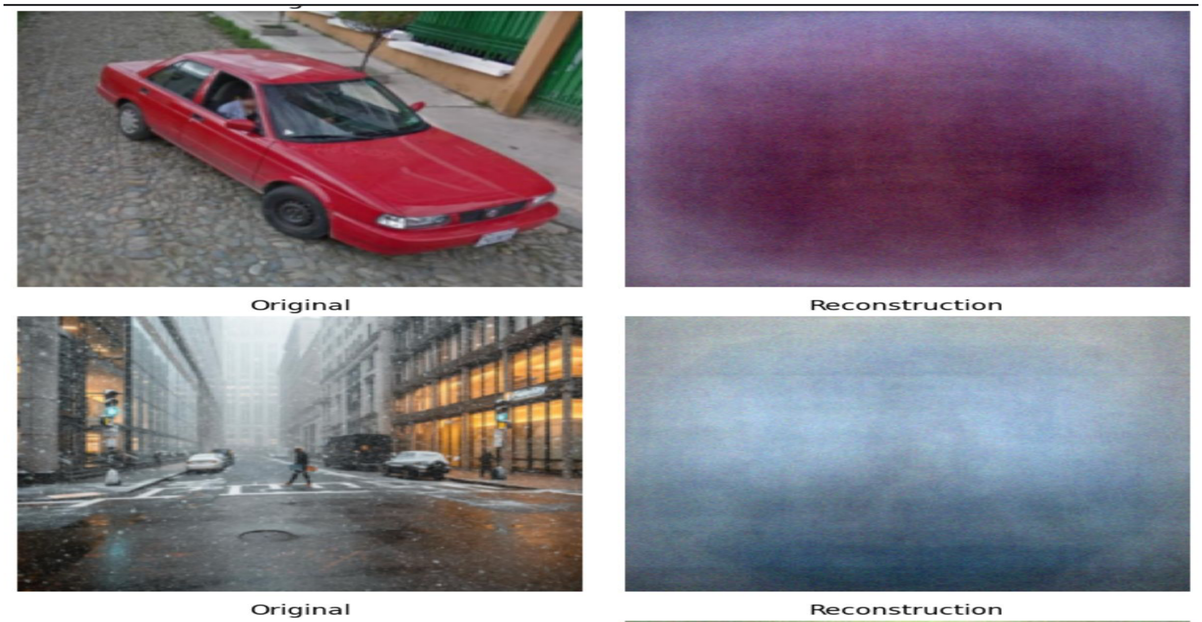


*Figure: failed reconstructions in weakly trained baseline MAE*

These baseline experiments revealed the limitations of short-duration self-supervised training. Although the baseline SimCLR model appeared to exhibit some cluster structure when embeddings were visualized on training images, this organization did not generalize well to validation data, indicating that the learned representations remained immature and insufficiently robust. The baseline MAE model performed even more poorly, producing reconstruction outputs that consisted primarily of blurred color gradients and patch artifacts with little recognizable scene structure. Together, these results suggested that neither model had converged to meaningful representations after only 10 epochs of training and motivated the adoption of larger architectures and substantially longer training schedules. Consequently, the baseline models serve primarily as reference points for evaluating the improvements achieved by the final models presented in this study.

## 4.2 Improved SimCLR Training

The final SimCLR model employed a ResNet-50 encoder and was trained for 100 epochs using the same NT-Xent contrastive loss. The projection head consisted of two fully connected layers with Batch Normalization and ReLU activations, producing 128-dimensional projection embeddings for contrastive optimization. For downstream evaluation, 2048-dimensional encoder features extracted before the projection head were used, as these provide richer semantic representations for clustering, retrieval, and classification tasks.

## 4.3 Improved MAE Training

The improved MAE model retained the ViT-Tiny encoder architecture but was trained for a substantially longer schedule of 120 epochs, allowing the reconstruction objective to drive convergence to meaningful representations. The ViT-Tiny encoder tokenizes $16 \times 16$ patches into 192-dimensional embeddings via a learnable linear projection, processes them through multiple self-attention layers, and produces patch-level features used by the decoder. The decoder is a lightweight transformer with fewer layers and a smaller embedding dimension, designed to reconstruct normalized pixel values for masked patches using MSE loss.

Latent representations for visualization and evaluation were extracted as the CLS token embedding or the mean-pooled patch embeddings from the encoder, depending on the analysis task.

## 4.4 Experimental Pipeline

The complete experimental workflow proceeded through four stages. In the Training stage, both SimCLR and MAE models were trained on the 19k-image training split under the configurations described above. In the Feature Extraction stage, encoder embeddings were extracted for all training images and L2-normalized before analysis. In the Visualization stage, embeddings were projected to two and three dimensions using t-SNE and UMAP, with parameters tuned for perplexity and number of neighbors to balance local and global structure preservation. In the Analysis stage, a suite of quantitative and qualitative evaluations was applied, including k-NN probing, SVD spectrum analysis, intraclass versus interclass similarity, reconstruction quality inspection, and saliency mapping.

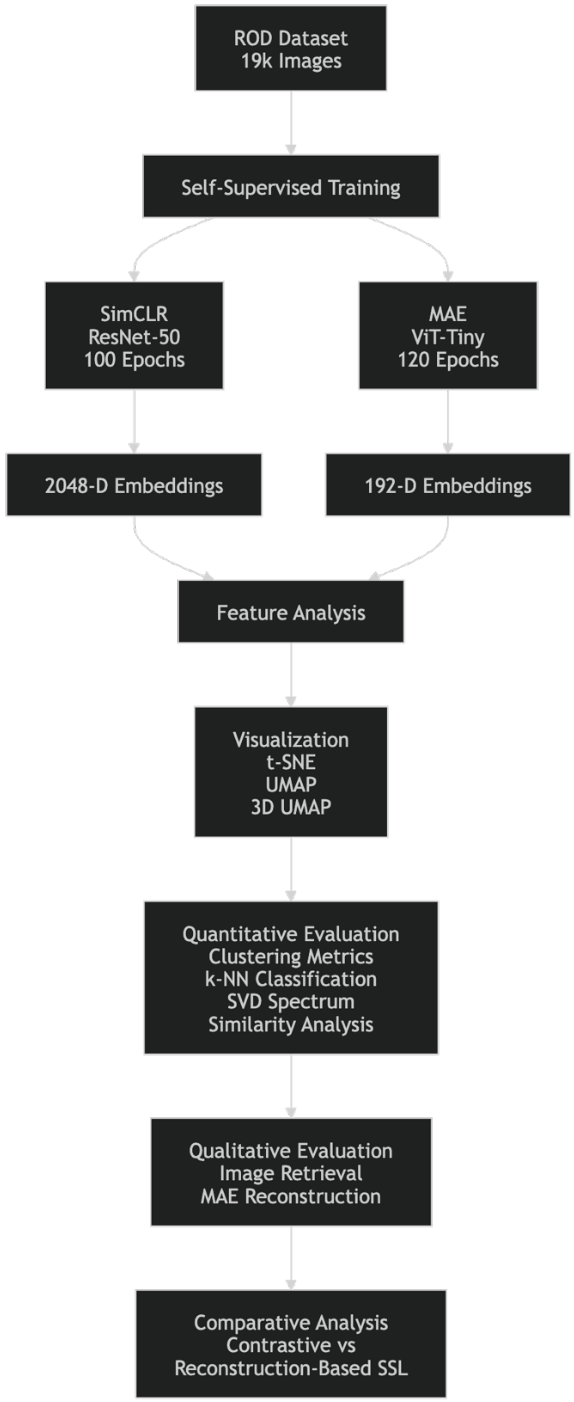


*Figure. Overview of the experimental pipeline. Models are trained on the obstacle detection dataset, embeddings are extracted and normalized, and a suite of visualization and evaluation techniques characterizes the learned representations.*

# 5. Experimental Setup

## 5.1 Hardware and Environment

All experiments were conducted in a Kaggle notebook environment utilizing GPU acceleration (NVIDIA T4 × 2). Training was performed under constrained session durations, which placed an upper bound on practical training length per run. The PyTorch deep learning framework was used throughout, with CUDA-accelerated data loading and mixed-precision training where applicable. All reported results correspond to the most complete and stable training runs achieved.

## 5.2 Training Configuration

*Table 1. Hyperparameter configuration for the improved SimCLR model.*

| Parameter | Value |
|---|---|
| Backbone | ResNet-50 |
| Projection Head | 2-layer MLP, 128-d output |
| Batch Size | 64 |
| Learning Rate | $1 \times 10^{-3}$ |
| Optimizer | Adam |
| Weight Decay | $1 \times 10^{-6}$ |
| Temperature (τ) | 0.5 |
| Training Epochs | 100 |
| Image Size | 224 × 224 |
| Loss Function | NT-Xent |

*Table 2. Hyperparameter configuration for the improved MAE model.*

| Parameter | Value |
|---|---|
| Encoder Architecture | ViT-Tiny |
| Patch Size | 16 × 16 |
| Encoder Embedding Dim. | 192 |
| Masking Ratio | 75% |
| Batch Size | 64 |
| Learning Rate | 1.5e-4 |
| Optimizer | AdamW |
| Weight Decay | 0.05 |
| Training Epochs | 150 |
| Image Size | 224 × 224 |
| Loss Function | Mean Squared Error (MSE) |

## 5.3 Evaluation Pipeline

Evaluation of learned representations followed a frozen-feature protocol: after pretraining, all encoder weights were frozen and embeddings were extracted for the full training split. These embeddings were then used as fixed feature vectors in all downstream analyses. For k-NN classification, a validation split was derived from labeled subsets of the dataset, enabling label-based accuracy evaluation without fine-tuning. The use of frozen features ensures that evaluation outcomes reflect the intrinsic quality of the SSL representations rather than task-specific adaptation.

# 6. Results and Analysis

This section presents the full suite of visualization and evaluation results for both the baseline and improved SimCLR and MAE models. For each analysis, the underlying methodology is described, axes and metrics are explained, baseline and improved models are compared, and an academic interpretation is provided. All figures referenced in this section should be inserted from the corresponding output files.

| | Model | Silhouette | Davies_Bouldin | Calinski_Harabasz |
|---|---|---|---|---|
| 0 | SimCLR | 0.08695 | 2.645251 | 60.437851 |
| 1 | MAE | -0.05999 | 4.146089 | 53.134468 |

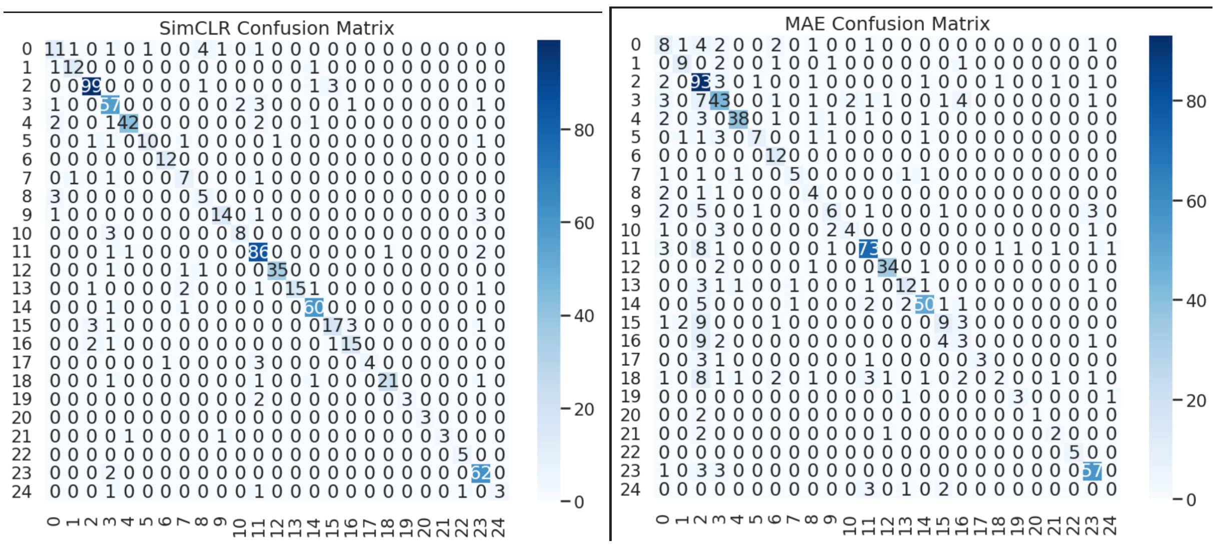


*Figure: Standard Metrics for the trained simCLR and MAE*

## 6.1 t-SNE and UMAP Visualization of Latent Embeddings

t-SNE and UMAP were used to visualize the learned embedding spaces of the improved SimCLR and MAE models. Each point represents an image and is colored according to its obstacle class label. The SimCLR embeddings form relatively compact and well-separated clusters, indicating strong semantic discrimination and effective class-level organization. In contrast, the MAE embeddings exhibit a more diffuse structure with greater overlap between classes, although meaningful local groupings remain visible. The UMAP projections reinforce these observations, showing clearer cluster separation for SimCLR and a smoother, more continuous manifold for MAE. Overall, the visualizations suggest that the contrastive objective of SimCLR produces more discriminative representations, while the reconstruction-based MAE objective captures broader contextual relationships between samples.

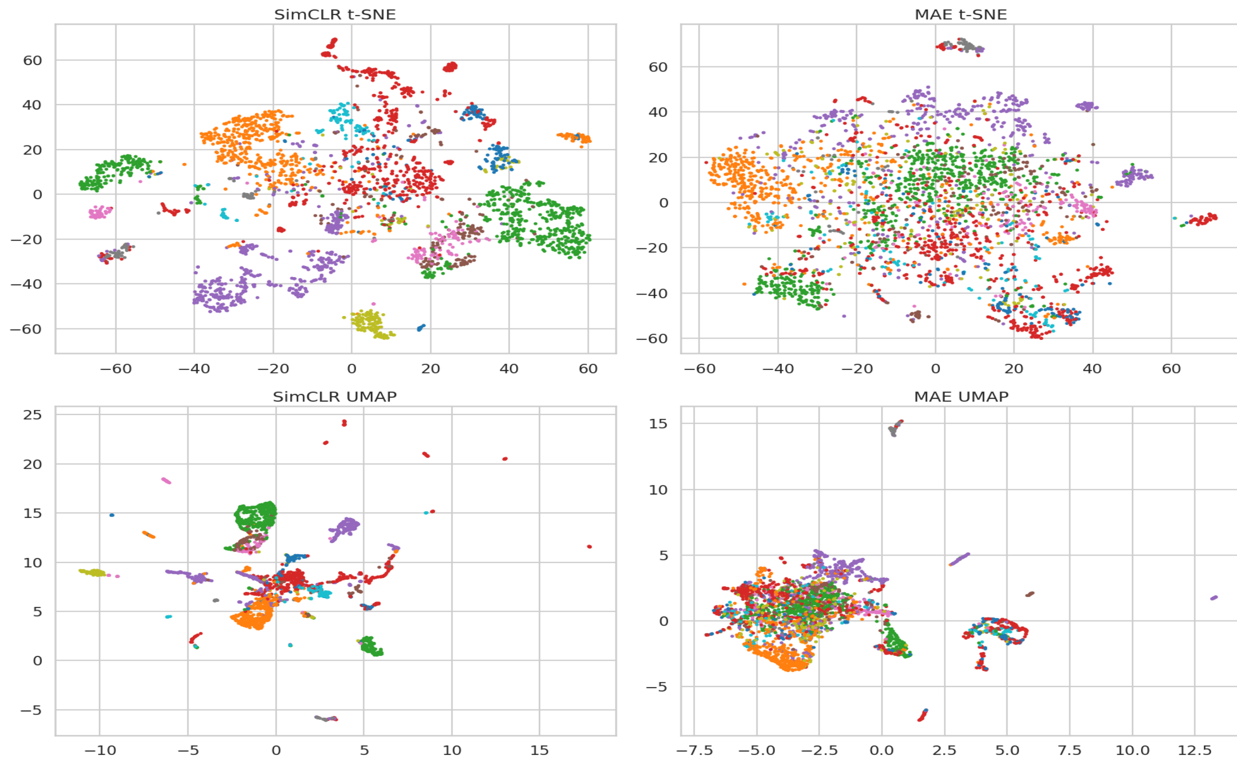


*Figure 5. t-SNE projections of latent embeddings for all four experimental conditions. Points are colored by ground-truth obstacle class. Improved models produce substantially more organized embedding distributions, with SimCLR-Improved exhibiting the tightest semantic clusters.*

t-SNE and UMAP were used to visualize the latent representations learned by the improved SimCLR and MAE models. Each point corresponds to a single image and is colored according to its obstacle class label. Although both methods project high-dimensional embeddings into two dimensions, t-SNE emphasizes local neighborhood preservation while UMAP provides additional insight into the global organization of the embedding space.

The SimCLR embeddings exhibit clear cluster formation in both t-SNE and UMAP projections, with several obstacle classes occupying compact and well-separated regions. This behavior indicates that the contrastive objective successfully organizes semantically similar samples into coherent neighborhoods while maintaining separation between different classes. The resulting latent space is highly discriminative and supports downstream classification and retrieval tasks.

In contrast, the MAE embeddings form a more continuous and overlapping manifold which are noticeably distributed across the space. While local groupings are still visible, class boundaries are less distinct and substantial overlap exists between several categories. This reflects the reconstruction-based learning objective, which encourages the model to capture contextual and structural information rather than explicitly maximizing inter-class separation. Nevertheless, meaningful semantic organization remains evident, indicating that the model has learned useful representations despite weaker clustering performance.

Overall, the visualization results suggest that SimCLR produces more discriminative and class-separable embeddings, whereas MAE learns smoother representations that prioritize contextual similarity over strict category separation.

## 6.3 Interactive 3D t-SNE and UMAP Analysis

To investigate latent geometry beyond what two-dimensional projections can convey, UMAP was also applied to produce three-dimensional embeddings, rendered as an interactive visualization. The additional dimension reveals depth and overlap structure that is inevitably collapsed in 2D projections: clusters that appear to overlap in a 2D plane may be well-separated along the third axis.

In the 3D UMAP visualization of SimCLR-Improved embeddings, cluster separability is further confirmed: most semantic groups occupy distinct volumetric regions with minimal interpenetration. For MAE-Improved, the 3D projection reinforces the observation of a smooth, low-curvature manifold with gradual semantic transitions. These observations are consistent with the theoretical predictions outlined in Section 2.5.

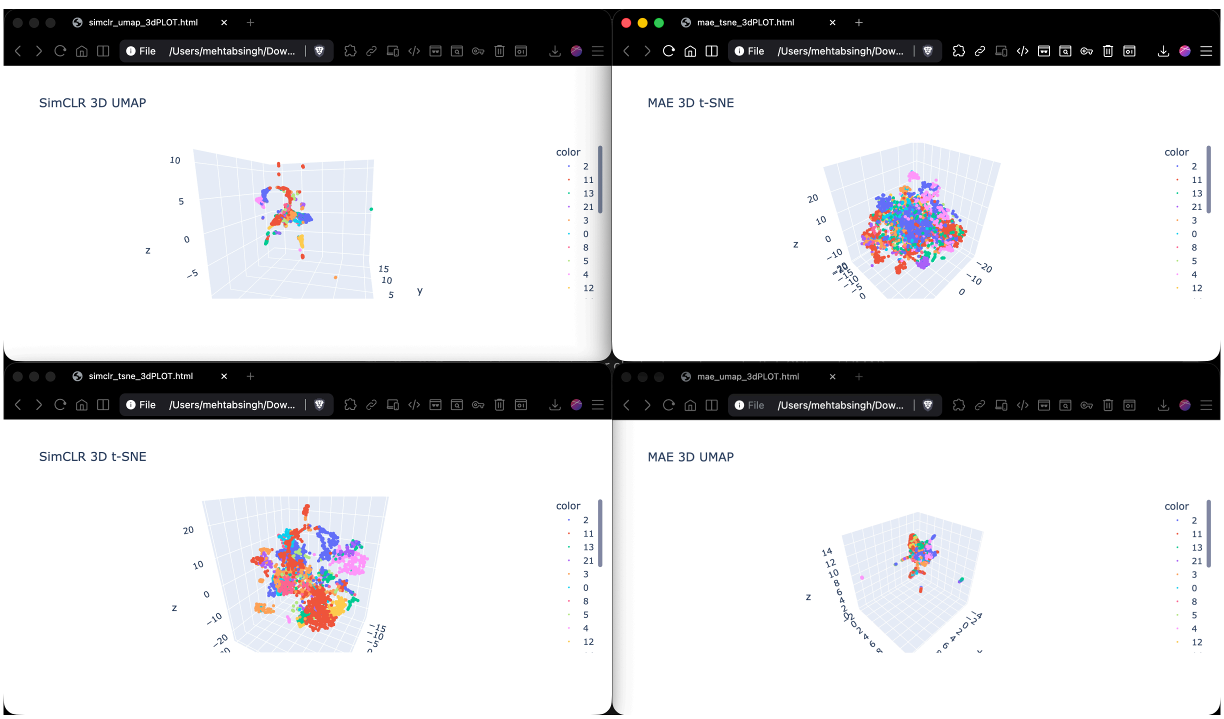


*Figure 7. Three-dimensional t-SNE and UMAP projection of improved model embeddings. The additional dimension reveals depth-separated cluster structure in SimCLR-Improved and smooth but distributed manifold topology in MAE-Improved. An interactive HTML version accompanies this report.*

## 6.4 k-NN Linear Evaluation

k-nearest-neighbor (k-NN) classification provides a non-parametric measure of representation quality. For each test image, the k most similar training embeddings (by cosine distance) are identified, and the majority class label among those neighbors is predicted. This metric evaluates the degree to which semantically similar images occupy neighboring regions in the embedding space, which is a direct measure of the discriminative utility of the learned representations.

| Model | Accuracy (%) | Precision | Recall | F1-Score |
|---|---|---|---|---|
| SimCLR | 86.63 | 86.55 | 78.37 | 80.99 |
| MAE | 68.71 | 65.08 | 55.23 | 57.23 |

*Table. k-NN classification accuracy for all four experimental conditions.*

The SimCLR embeddings achieved an accuracy of 86.6%, substantially outperforming the MAE embeddings, which achieved 68.7% accuracy. Similar trends were observed for precision, recall, and F1-score, with SimCLR consistently producing stronger results across all metrics. These findings indicate that the contrastive objective learned a more discriminative feature space in which samples belonging to the same obstacle category were more effectively grouped together.

The superior k-NN performance of SimCLR is consistent with the t-SNE and UMAP visualizations, clustering metrics, and confusion matrices, all of which showed clearer class separation and stronger semantic organization. Although MAE learned meaningful representations capable of supporting classification and retrieval, its reconstruction-based objective produced a less discriminative embedding space, resulting in lower nearest-neighbor classification performance.

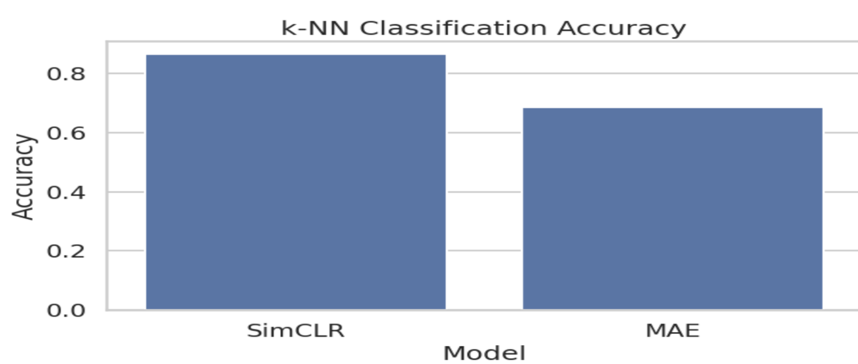


*Figure 8. k-NN classification accuracy for baseline and improved SimCLR and MAE models. Accuracy is computed on a held-out validation split using frozen encoder features. Higher accuracy indicates more linearly separable and semantically organized embeddings.*

## 6.5 SVD Eigenvalue Spectrum Analysis

Singular Value Decomposition (SVD) was applied to the embedding matrices to examine how information is distributed across latent dimensions. A gradual decay of singular values indicates that the representation utilizes many dimensions effectively, whereas a sharp decay suggests that information is concentrated in a smaller number of dominant directions.

The SVD spectra reveal a substantial difference between the two self-supervised learning approaches. SimCLR exhibits a gradual decline in singular values and achieves an effective rank of 89.9, indicating that information is distributed across a large portion of the embedding space. In contrast, the MAE spectrum decays much more rapidly, with an effective rank of 19.4, suggesting that the learned representations occupy a lower-dimensional subspace.

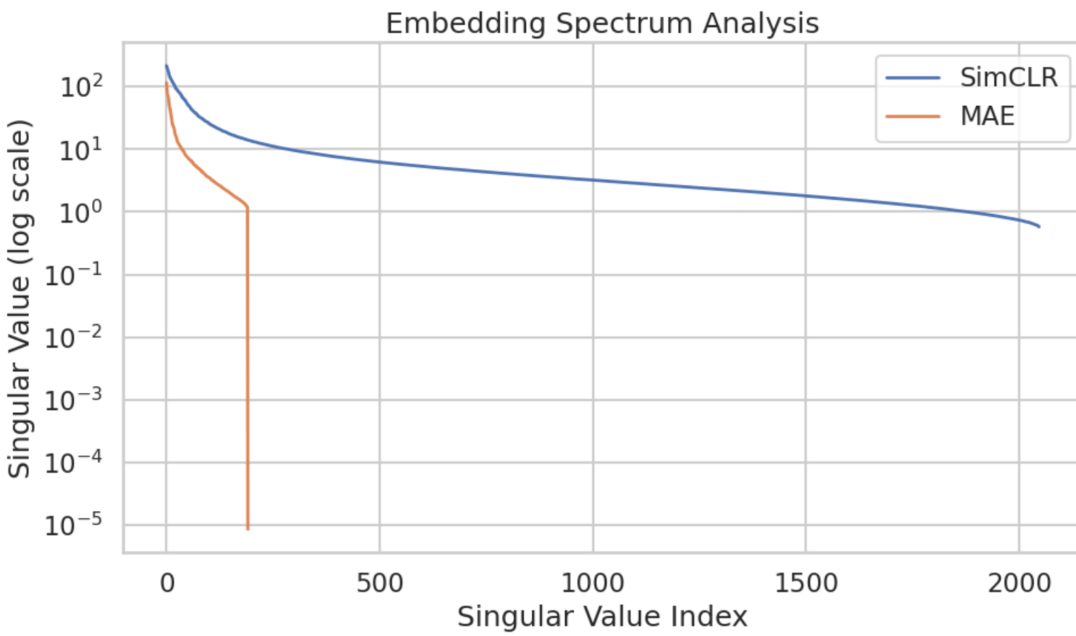


*Figure 9. Singular value spectra of the embedding matrices for all four experimental conditions. Gradual spectral decay indicates high effective rank and diverse feature utilization. Sharp early decay indicates potential dimensional collapse.*

These findings indicate that SimCLR learns richer and more diverse feature representations, which is consistent with its stronger clustering performance, higher k-NN accuracy, and clearer class separation observed in the visualization results. Although MAE learns meaningful semantic features, its reconstruction-based objective produces a more compact latent structure with lower representational diversity.

## 6.6 Intraclass vs. Interclass Similarity Analysis

To quantify semantic organization in the embedding space, pairwise cosine similarities were computed for samples belonging to the same class (intraclass) and for samples belonging to different classes (interclass). A well-structured representation should exhibit high intraclass similarity and lower interclass similarity, producing a clear separation between the two distributions.

For both models, intraclass similarities are generally higher than interclass similarities, confirming that semantically related samples are grouped together in the learned feature space. However, the separation between the two distributions is noticeably stronger for SimCLR. The SimCLR embeddings show a tighter interclass distribution centered near zero and a distinct shift toward higher similarity values for intraclass pairs, indicating stronger class discrimination.

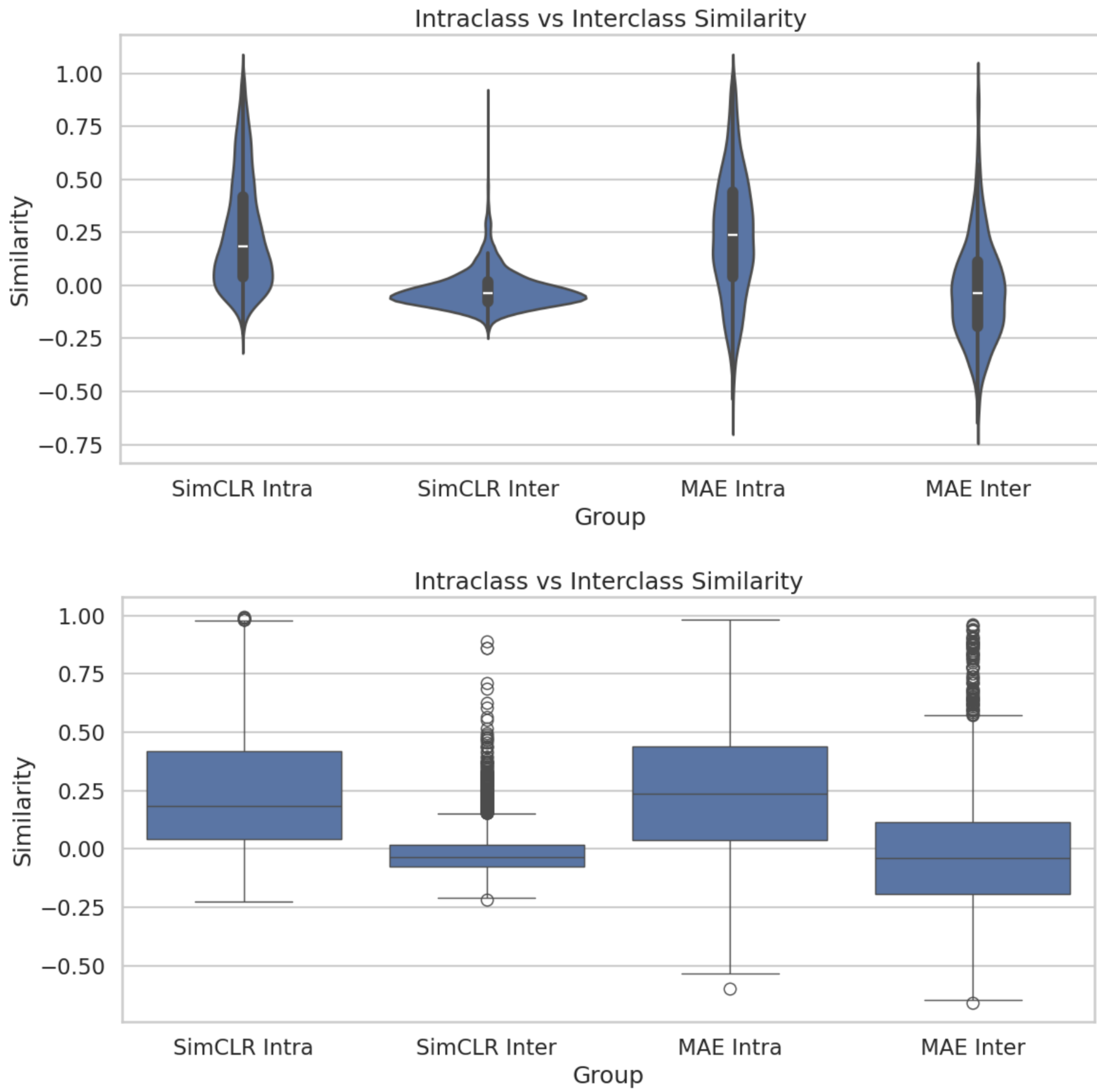


*Figure 10. Distribution of pairwise cosine similarities within classes (intraclass) and between classes (interclass) for the improved models. A larger gap between the distributions indicates better class separability in the embedding space.*

The MAE embeddings also demonstrate meaningful semantic organization, but the distributions exhibit greater overlap. This behavior suggests that the reconstruction objective encourages broader contextual similarity rather than strict category-level separation. These observations are consistent with the clustering metrics, k-NN evaluation, and visualization results, all of which indicate that SimCLR produces a more discriminative embedding space while MAE learns smoother semantic representations.

## 6.7 MAE Reconstruction Analysis

A key diagnostic for MAE is the visual quality of its reconstructions. Reconstruction quality provides evidence of whether the encoder has learned sufficient contextual and structural information to support pixel-level prediction, and serves as a direct measure of the extent to which training has converged.

Reconstruction quality was evaluated to assess the extent to which the MAE model learned meaningful visual representations from heavily masked inputs. Figure 11 presents representative examples showing the original image, the masked input, and the reconstructed output produced by the model.

The reconstruction results are relatively coarse and fail to recover many fine-grained image details. While the model is able to preserve basic scene structure, approximate object locations, and dominant color distributions, the reconstructed images remain blurry and contain visible patch artifacts. Object boundaries, textures, and high-frequency visual features are often poorly reconstructed, indicating that the model did not fully converge to a high-fidelity reconstruction solution.

Several factors likely contributed to this behavior. First, the encoder used a relatively small configuration with an embedding dimension of 192 and 12 transformer blocks, substantially smaller than the ViT-Base and ViT-Large architectures commonly used in MAE literature. Second, a masking ratio of 75% leaves only 25% of image patches visible to the encoder, making reconstruction particularly challenging. Finally, computational and time constraints limited the scale of experimentation, preventing the use of larger architectures and longer training schedules that are typically required for high-quality MAE reconstructions.

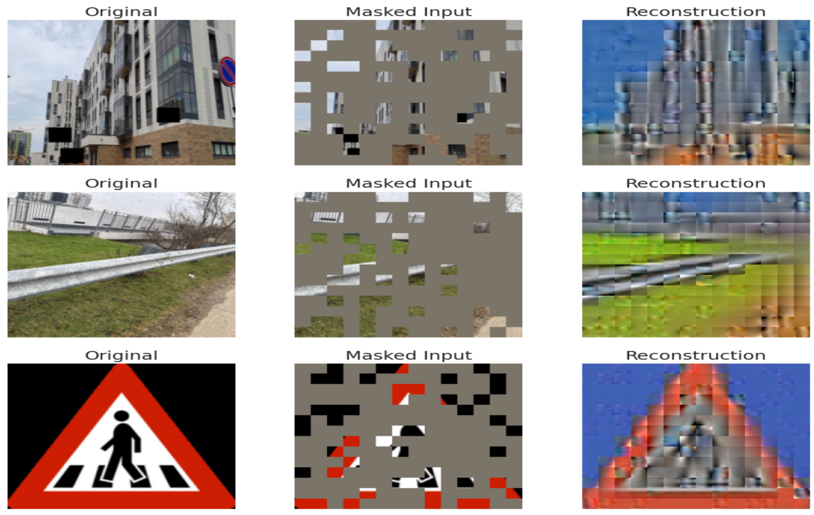


*Figure:. MAE reconstruction examples showing the original image, masked input, and reconstructed output. Although the model captures coarse scene structure and color information, reconstruction quality remains limited due to the compact architecture, aggressive masking ratio, and computational constraints of the training setup.*

Despite the limited reconstruction fidelity, the model still learned useful visual representations. The latent embeddings demonstrated meaningful clustering behavior, non-trivial retrieval performance, and a k-NN accuracy of 68.7%, indicating that representation quality and reconstruction quality do not necessarily improve at the same rate. This observation highlights an important characteristic of reconstruction-based self-supervised learning: a model may learn semantically useful features even when pixel-level reconstruction remains imperfect.

## 6.8 Image Retrieval Analysis

The SimCLR embeddings produced consistently strong retrieval results across multiple obstacle categories. Query images containing benches, stairs, chairs, and buses typically returned visually and semantically similar examples, demonstrating that the contrastive objective learned highly discriminative representations. Retrieved samples generally belonged to the same obstacle category despite variations in viewpoint, lighting conditions, and background context.

The MAE retrieval results were noticeably weaker. In many cases, the retrieved images did not correspond to the exact obstacle category present in the query image. However, the model frequently returned images with similar scene structure and contextual content. For example, road scenes tended to retrieve other road scenes, indicating that the MAE encoder successfully captured global visual context even when category-level discrimination was limited. This behavior is consistent with the reconstruction objective, which encourages learning broad structural and contextual relationships rather than explicitly separating semantic classes.

Overall, the retrieval experiments further support the quantitative findings. SimCLR learned a more discriminative embedding space that is well suited for retrieval and recognition tasks, whereas MAE produced representations that capture scene-level context but provide weaker category-level separation.

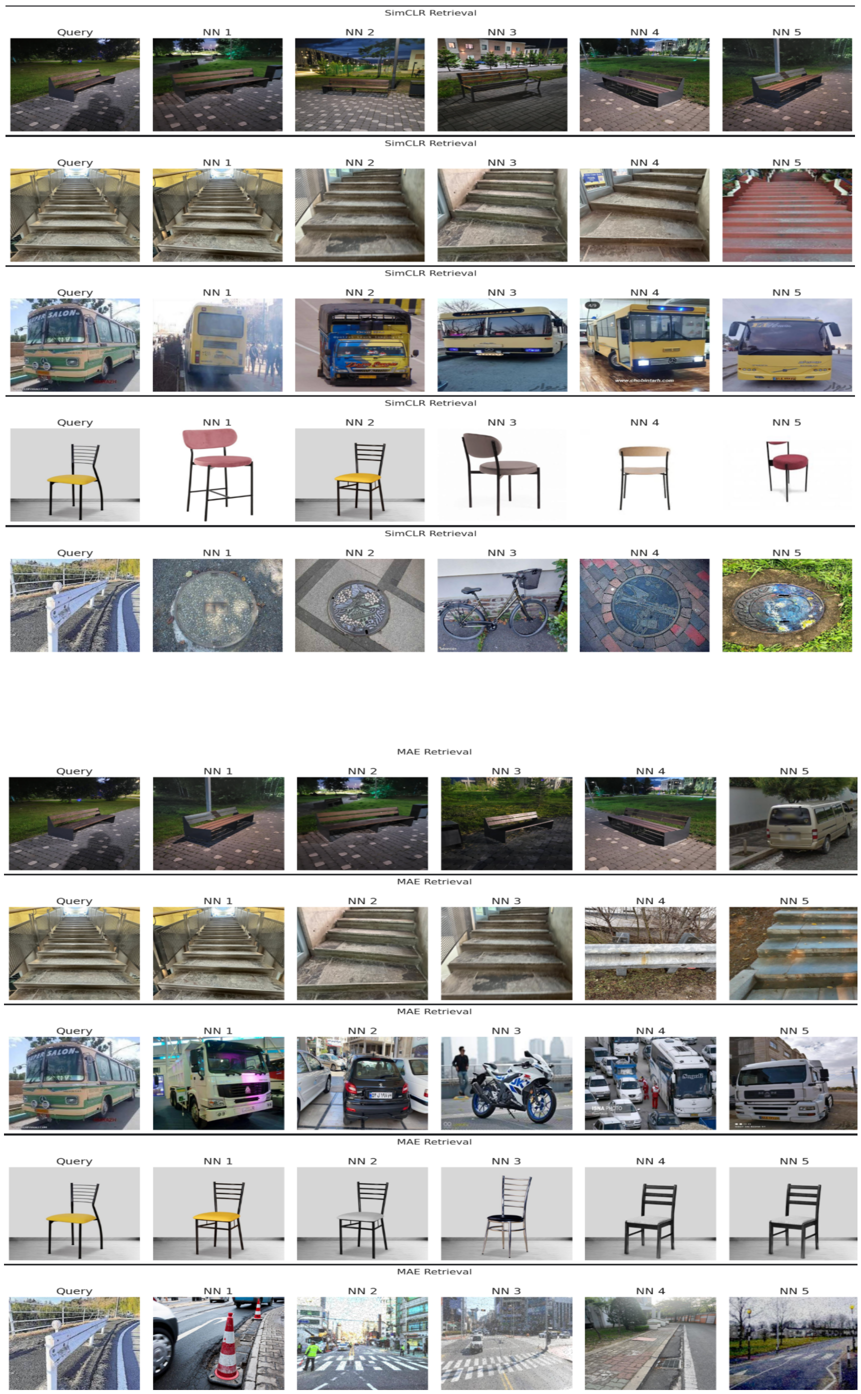


*Figure: Image retrieval results using cosine similarity in the learned embedding space. For each query image, the nearest neighbors are retrieved from the validation set. SimCLR consistently retrieves semantically similar obstacle categories, while MAE retrieval is influenced more strongly by contextual and structural similarity.*

# 7. Discussion

The results demonstrate clear differences between contrastive and reconstruction-based self-supervised learning approaches. Across nearly all quantitative and qualitative evaluations, SimCLR produced stronger and more discriminative representations than MAE. The superiority of SimCLR was consistently reflected in the clustering metrics, visualization results, k-NN classification performance, retrieval experiments, and SVD spectrum analysis.

The t-SNE and UMAP visualizations revealed that SimCLR learned compact and well-separated clusters corresponding to obstacle categories, while MAE produced a smoother and more overlapping latent manifold. These observations were supported quantitatively by the clustering metrics, where SimCLR achieved a higher Silhouette Score (0.087), lower Davies–Bouldin Index (2.65), and higher Calinski–Harabasz Score (60.44) than MAE. Together, these results indicate stronger class separability and more coherent latent organization in the contrastive embedding space.

The downstream evaluation further reinforced these findings. SimCLR achieved a k-NN classification accuracy of 86.6%, substantially outperforming MAE's 68.7% accuracy. The confusion matrices showed stronger diagonal dominance for SimCLR, indicating that images belonging to the same obstacle category occupied consistent neighborhoods in the latent space. Similarly, the image retrieval experiments demonstrated that SimCLR reliably retrieved semantically identical obstacle classes, whereas MAE retrieval was influenced more strongly by scene context and overall image structure.

The SVD analysis provided additional insight into representation quality. SimCLR achieved an effective rank of approximately 89.9 compared with 19.4 for MAE, indicating significantly greater utilization of the embedding space and a richer distribution of information across latent dimensions. This suggests that the contrastive objective encourages more diverse and discriminative feature representations.

Although MAE underperformed relative to SimCLR, the results should not be interpreted as a failure of reconstruction-based learning. Despite weaker clustering and retrieval performance, MAE still learned meaningful visual representations capable of supporting classification, retrieval, and semantic grouping. The model successfully captured global scene structure and contextual information, as evidenced by both the embedding visualizations and retrieval results. However, reconstruction quality remained relatively poor, with blurry outputs and limited recovery of fine-grained image details. This is likely attributable to the compact architecture, aggressive masking ratio, and limited computational resources available during training.

Overall, the findings indicate that contrastive learning is particularly effective when strong semantic discrimination is required, whereas reconstruction-based learning produces broader contextual representations that may be beneficial for tasks emphasizing scene understanding rather than category separation.

# 8. Limitations and Future Work

## 8.1 Current Limitations

Several limitations should be considered when interpreting the results of this study. First, computational constraints limited the scale of experimentation. The models were trained using freely available cloud resources, restricting the size of architectures, batch sizes, and training duration that could be explored. In particular, the MAE implementation employed a relatively small Vision Transformer with an embedding dimension of 192, substantially smaller than the architectures commonly used in large-scale MAE studies.

Second, formal ablation studies were not conducted. Evaluating the individual contributions of architectural choices, masking ratios, embedding dimensions, projection heads, and training hyperparameters would require retraining multiple model variants, which was not feasible within the project timeframe. Consequently, the analysis focused on a comprehensive comparison between optimized SimCLR and MAE implementations rather than component-level investigations.

Third, evaluation was performed on a single obstacle detection dataset. Although the dataset contains meaningful visual diversity, additional experiments on larger and more varied benchmarks would provide stronger evidence regarding the generalizability of the learned representations.

## 8.2 Future Work

**8.2 Future Work**

The most impactful near-term extension of this work would be to scale both architectures. Replacing ResNet-50 with a larger ResNet variant or a modern convolution-based architecture, and replacing ViT-Tiny with ViT-Small or ViT-Base, would substantially increase the ceiling of achievable representation quality and bring the experimental setting closer to the conditions under which published benchmarks are established.

A second important direction is expanding the evaluation protocol to include linear probing and full fine-tuning on held-out datasets. This would provide a cl From a methodological perspective, it would be valuable to compare SimCLR and MAE against more recent SSL methods including BYOL (Bootstrap Your Own Latent), DINO (self-DIstillation with NO labels), and I-JEPA (Image Joint Embedding Predictive Architecture). These methods represent distinct points in the design space of SSL objectives and may provide different representational tradeoffs. DINO in particular has shown strong performance on local feature tasks without requiring negative pairs, making it an interesting intermediate between pure contrastive and reconstruction-based approaches.

Future work could address these limitations by training larger MAE architectures, extending training duration, and exploring alternative masking strategies. Additional downstream evaluations, including linear probing, object detection, and semantic segmentation, would provide a more complete assessment of representation quality. Formal ablation studies could further identify the architectural and optimization choices that contribute most strongly to performance. Finally, scaling the experiments to larger datasets and more powerful computational resources would enable a deeper comparison between contrastive and reconstruction-based self-supervised learning approaches.

Finally, the present study treats SimCLR and MAE as separate systems. A natural extension is to investigate hybrid training objectives for example, combining a contrastive loss on global embeddings with a reconstruction loss on local patches to determine whether such objectives can simultaneously produce discriminative and structurally coherent representations. Joint or sequential pretraining schedules that first optimize one objective and then fine-tune with the other are also worth exploring.

# 9. Conclusion

This project investigated and compared two prominent self-supervised learning paradigms, SimCLR and Masked Autoencoders (MAE), for representation learning on an obstacle detection dataset. Both approaches successfully learned useful visual representations without requiring manual annotation during training, demonstrating the practical potential of self-supervised learning for computer vision tasks.

Comprehensive evaluation using dimensionality-reduction visualizations, clustering metrics, k-NN classification, similarity analysis, image retrieval, reconstruction assessment, and spectral analysis revealed a consistent performance advantage for SimCLR. The contrastive model produced highly discriminative embeddings characterized by strong cluster separation, superior retrieval quality, higher downstream classification accuracy, and greater representational diversity. SimCLR achieved an accuracy of 86.6% in k-NN evaluation and outperformed MAE across all clustering metrics and embedding quality measures.

The MAE model learned meaningful contextual representations and demonstrated non-trivial retrieval and classification capabilities. However, the learned embeddings exhibited weaker class separation, lower effective rank, and reduced downstream performance. Reconstruction quality remained below expectations, likely due to the compact architecture and computational limitations of the training setup. Nevertheless, the results confirmed that useful semantic representations can emerge even when reconstruction fidelity is limited.

Overall, the findings suggest that contrastive learning is better suited for applications requiring strong semantic discrimination and retrieval performance, while reconstruction-based learning offers a complementary approach focused on contextual understanding and structural representation. The study highlights the strengths and trade-offs of both paradigms and provides practical insight into their behavior under realistic computational constraints.

# 10. References

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